

2017 HKDSE Physics

Paper 2 — Elective Sections

Original questions and official answers with detailed English reasoning. Calculations show relationships, substitution and units; each structured subpart is presented separately.

Section A: Astronomy and Space Science

Q.1: Multiple-choice questions

Question 1.1

Normally, Mars moves eastward in the sky throughout the year. In May and June 2016, retrograde motion of Mars was observed. Which of the following statements is/are correct?

- (1) During that period, Mars moved westward in the sky.
- (2) Retrograde motion is observed because the Earth moves faster than Mars.
- (3) Ptolemy's geocentric model cannot explain the retrograde motion.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: C

Retrograde motion means an apparent westward motion against the background stars. Earth moves faster in its inner orbit and overtakes Mars, producing this apparent reversal. Ptolemy's geocentric model could reproduce retrograde loops using epicycles, so (3) is false. Answer: C.

Question 1.2

Two astronauts are experiencing “weightlessness” in a space station. The masses of the astronauts are 50 kg and 70 kg respectively. Which of the following statements is/are correct?

- (1) No gravitational force is acting on the two astronauts by the Earth.
- (2) The net forces acting on the two astronauts are the same.
- (3) The two astronauts have the same acceleration.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: B

The astronauts and station are in common free fall; weightlessness means no supporting force is felt, not that gravity vanishes. At the same orbital position, their accelerations are the same, but $F = ma$ gives different forces for different masses. Only (3) is correct. Answer: B.

Question 1.3

It is given that the Moon's radius is $0.273R$, where R is the radius of the Earth. The acceleration due to gravity on the Moon's surface is $(1/6)g$, where g is the acceleration due to gravity on the Earth's surface. If v is the escape velocity from the surface of the Earth, what is the escape velocity from the surface of the Moon?

- A. $0.046v$
- B. $0.167v$
- C. $0.213v$
- D. $0.273v$

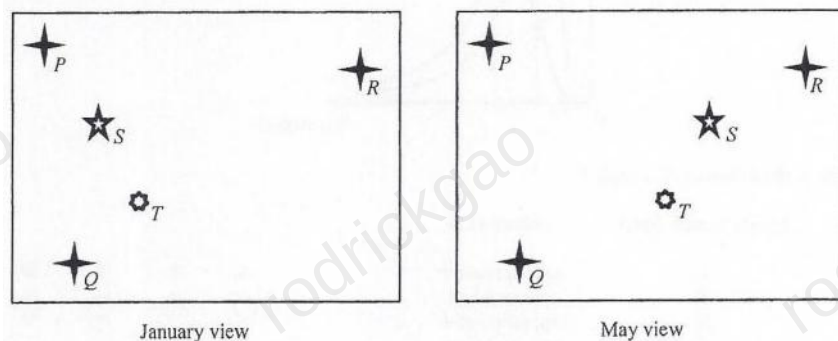
Official answer: C

Escape speed is $v_e = \sqrt{2GM/R} = \sqrt{2gR}$. Thus $v_{\text{Moon}}/v_{\text{Earth}} = \sqrt{[(1/6)(0.273)]} = 0.213$. The Moon's escape speed is $0.213v$. Answer: C.

Question 1.4

The following shows two pictures of the same region of the sky taken in January and May of a certain year. P, Q, R, S and T are five stars. Which of the following statements **MUST BE** correct?

- (1) Stars P, Q and R are equidistant from the Earth.
- (2) The parallax of star S is smaller than that of star T.
- (3) Star S is closer to the Earth than star T.



- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: B

Relative to the almost fixed background stars P, Q and R, S shows the larger displacement between the two views. A larger annual parallax indicates a smaller distance, so S is closer than T. Unresolved shifts of the background stars do not prove that they are equidistant. Only (3) is required. Answer: B.

Question 1.5

The table below shows the apparent magnitudes and absolute magnitudes of three stars. Which of the following is correct?

star	apparent magnitude	absolute magnitude
Sirius A	-1.47	1.42
Vega	0.03	0.58
Polaris A	1.98	-3.64

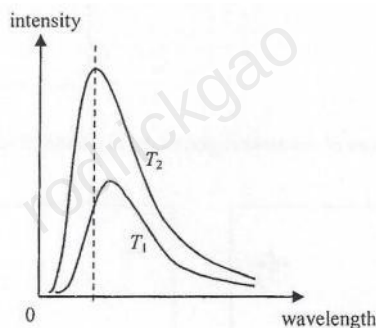
- A. Greatest luminosity: Sirius A; farthest from Earth: Sirius A; brightest as seen from Earth: Polaris A
- B. Greatest luminosity: Sirius A; farthest from Earth: Vega; brightest as seen from Earth: Sirius A
- C. Greatest luminosity: Polaris A; farthest from Earth: Vega; brightest as seen from Earth: Polaris A
- D. Greatest luminosity: Polaris A; farthest from Earth: Polaris A; brightest as seen from Earth: Sirius A

Official answer: D

Smaller absolute magnitude means greater luminosity, so Polaris A (-3.64) is the most luminous. From $m - M = 5 \log_{10}(d/10 \text{ pc})$, Polaris A has the largest distance modulus, $1.98 - (-3.64) = 5.62$, and is farthest. Sirius A has the smallest apparent magnitude, -1.47, so it appears brightest. Answer: D.

Question 1.6

The diagram shows the spectra of radiation from a black body at two different temperatures, T_1 and T_2 . Which of the following is correct?



- A. Higher temperature: T_1 ; colour at T_2 appears redder
- B. Higher temperature: T_1 ; colour at T_2 appears bluer
- C. Higher temperature: T_2 ; colour at T_2 appears redder
- D. Higher temperature: T_2 ; colour at T_2 appears bluer

Official answer: D

The T_2 spectrum peaks at a shorter wavelength and has greater intensity. Wien's law $\lambda_{\text{max}} T = \text{constant}$ therefore gives $T_2 > T_1$. Its increased short-wavelength contribution makes it appear bluer. The original option combining these two statements is D. Answer: D.

Question 1.7

It is known that the Sun is a class G star, and the star Zeta Puppis is a class O supergiant. Given that the sequence of spectral classes is O B A F G K M, which of the following is correct?

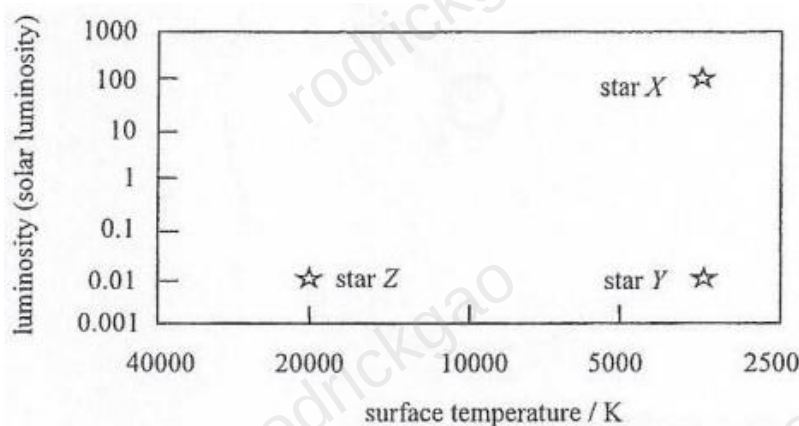
- A. Higher surface temperature: Zeta Puppis; greater luminosity: Zeta Puppis
- B. Higher surface temperature: Zeta Puppis; greater luminosity: the Sun
- C. Higher surface temperature: the Sun; greater luminosity: Zeta Puppis
- D. Higher surface temperature: the Sun; greater luminosity: the Sun

Official answer: A

Spectral classes O B A F G K M are ordered from hotter to cooler, so an O star has higher surface temperature than the G-class Sun. Zeta Puppis is also a supergiant with far greater radiating area; $L = 4\pi R^2 \sigma T^4$ gives greater luminosity. Answer: A.

Question 1.8

The figure shows some information of stars X, Y and Z. Which of the following comparisons about the size of the three stars is correct?



- A. $X > Y > Z$
- B. $X = Y > Z$
- C. $X > Y = Z$
- D. $Z > Y > X$

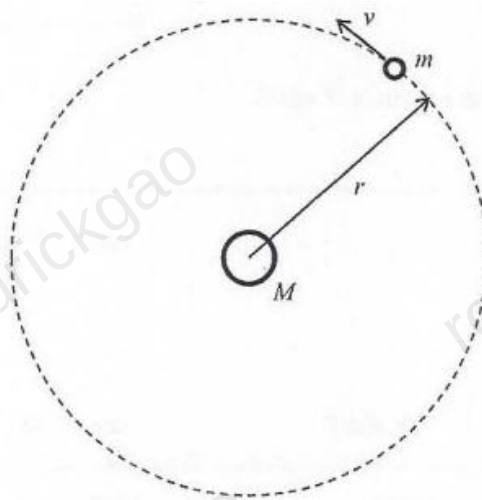
Official answer: A

From $L = 4\pi R^2 \sigma T^4$, R is proportional to $\sqrt{L/T^4}$. X and Y have the same temperature but X is much more luminous, so X is larger. Y and Z have the same luminosity but Y is cooler, so Y is larger. Hence $X > Y > Z$. Answer: A.

Q.1: Structured question

(a) Figure 1.1 shows an object of mass m orbiting around a star of mass M with a radius of r . The velocity of the object is v .

Figure 1.1



(i) Using Newton's law of gravitation, show that $v^2 = GM/r$, where G is the universal gravitational constant. (1 mark)

(ii) Hence, or otherwise, show that $T^2 = (4\pi^2/GM)r^3$, where T is the period of the motion of the object. (2 marks)

(b) Stars and gases orbit around the centre of the M33 Galaxy. At a position X near the edge of the galaxy (3.98×10^{20} m from the centre of the galaxy), the orbital velocity of the hydrogen gas is about 1.23×10^5 m s⁻¹. You may assume that the hydrogen gas at X orbits with a circular orbit.

(i) One of the spectral lines of hydrogen gas (the H I line) has a wavelength of 21.106 cm. If the hydrogen gas at X is moving towards the Earth along the line of sight, what would be the observed wavelength of the H I line? (2 marks)

(ii) How long would it take for the hydrogen gas at X to complete one orbit around the M33 Galaxy? (1 mark)

(iii) Using the results of (a)(ii), or otherwise, estimate the mass of the M33 Galaxy in solar mass.

Given: 1 AU = 1.50×10^{11} m, and 1 year = 3.16×10^7 s. (3 marks)

(iv) Astronomers estimated that the total mass of luminous objects in the M33 Galaxy is 7×10^9 solar mass. Compare this to your answer in (b)(iii) and suggest a reason to explain the difference, if any. (1 mark)

Official marking scheme

Solution	Marks	Remarks
1. (a) (i) $\frac{GMm}{r^2} = \frac{mv^2}{r}$ $v^2 = \frac{GM}{r}$	1M	
	1	
(ii) $T = \frac{2\pi r}{v}$ $T^2 = \frac{4\pi^2 r^2}{v^2}$ $= \frac{4\pi^2 r^2}{\left(\frac{GM}{r}\right)}$ from (i) $= \frac{4\pi^2}{GM} r^3$	1M 1M	
	2	
(b) (i) By $\frac{\Delta\lambda}{\lambda_0} \approx \frac{v}{c}$ $\Delta\lambda \approx \frac{v}{c} \lambda_0 = \frac{1.23 \times 10^5}{3 \times 10^8} \times 21.106$ $= 8.65346 \times 10^{-3} \text{ cm}$ $\lambda = \lambda_0 - \Delta\lambda$ $= 21.106 - 8.65346 \times 10^{-3}$ $= 21.097 \text{ cm}$	1M 1A	
	2	
(ii) $T = \frac{2\pi r}{v}$ $= \frac{2 \times 3.14 \times (3.98 \times 10^{20})}{1.23 \times 10^5}$ $= 2.03 \times 10^{16} \text{ s (or } 6.42 \times 10^8 \text{ yr)}$	1A	
	1	

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Official marking scheme

Solution	Marks	Remarks
1. (a) (i) $\frac{GMm}{r^2} = \frac{mv^2}{r}$ $v^2 = \frac{GM}{r}$	1M	
	1	
(ii) $T = \frac{2\pi r}{v}$ $T^2 = \frac{4\pi^2 r^2}{v^2}$ $= \frac{4\pi^2 r^2}{\left(\frac{GM}{r}\right)}$ from (i) $= \frac{4\pi^2}{GM} r^3$	1M 1M	
	2	
(b) (i) By $\frac{\Delta\lambda}{\lambda_0} \approx \frac{v}{c}$ $\Delta\lambda \approx \frac{v}{c} \lambda_0 = \frac{1.23 \times 10^5}{3 \times 10^8} \times 21.106$ $= 8.65346 \times 10^{-3} \text{ cm}$ $\lambda = \lambda_0 - \Delta\lambda$ $= 21.106 - 8.65346 \times 10^{-3}$ $= 21.097 \text{ cm}$	1M 1A	
	2	
(ii) $T = \frac{2\pi r}{v}$ $= \frac{2 \times 3.14 \times (3.98 \times 10^{20})}{1.23 \times 10^5}$ $= 2.03 \times 10^{16} \text{ s (or } 6.42 \times 10^8 \text{ yr)}$	1A	
	1	

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(a)(ii) For uniform circular motion, $T = 2\pi r/v$. Thus $T^2 = 4\pi^2 r^2/v^2 = (4\pi^2/GM)r^3$.

(b)(ii) $T = 2\pi r/v = 2\pi(3.98 \times 10^{20})/(1.23 \times 10^5) = 2.03 \times 10^{16} \text{ s} \approx 6.43 \times 10^8 \text{ years.}$

(b)(iii) Compare with Earth's orbit: $M/M_S = (r/1 \text{ AU})^3(1 \text{ year}/T)^2$. Using unrounded T gives $(3.98 \times 10^{20}/1.50 \times 10^{11})^3(3.16 \times 10^7/T)^2 \approx 4.51 \times 10^{10}$ solar masses. Using the rounded $T = 2.03 \times 10^{16} \text{ s}$ gives 4.53×10^{10} , as in the marking scheme.

(b)(iv) This dynamical estimate is about 6.5 times the luminous mass of 7×10^9 solar masses. Additional non-luminous matter, such as dark matter, contributes gravity without appearing in the luminous-mass estimate.

Section B: Atomic World

Q.2: Multiple-choice questions

Question 2.1

Which of the following statements are correct according to Rutherford's atomic model?

- (1) Almost all the mass of an atom are concentrated at the nucleus.
- (2) Almost all the charges of an atom are concentrated at the nucleus.
- (3) Electrons orbit around the nucleus.

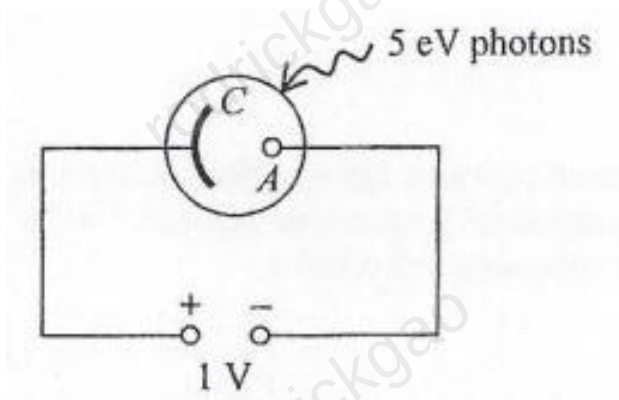
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: B

Rutherford's model concentrates almost all atomic mass and all positive charge in a small nucleus, with electrons orbiting outside it. Statement (2) incorrectly includes the negative charges of those electrons. Hence (1) and (3) are correct. Answer: B.

Question 2.2

A photocell is connected to a 1 V d.c. source as shown. A monochromatic light beam with each photon of energy 5 eV is incident on cathode C of the photocell so that photoelectrons are emitted. If the work function of cathode C is 2 eV, what is the maximum kinetic energy of the photoelectrons reaching anode A?



- A. 2 eV
- B. 3 eV
- C. 4 eV
- D. 6 eV

Official answer: A

At emission, $K_{\max} = hf - \phi = 5 - 2 = 3$ eV. The anode is 1 V below the cathode, so the field retards the electrons and reduces their kinetic energy by 1 eV. They arrive with maximum kinetic energy 2 eV. Answer: A.

Question 2.3

When monochromatic light of wavelengths λ and $(3/4)\lambda$ are incident on the cathode surface of a photocell separately, the stopping potentials are in the ratio of 1 : 2. What is the longest wavelength of monochromatic light that can cause photoelectrons to be emitted from the photocell?

- A. λ
- B. $(4/3)\lambda$
- C. $(3/2)\lambda$
- D. $(5/3)\lambda$

Official answer: C

Let $E = hc/\lambda$ and let the first stopping potential be V . Einstein's equation gives $eV = E - \phi$ and $2eV = 4E/3 - \phi$. Subtracting gives $eV = E/3$, hence $\phi = 2E/3$. The threshold wavelength is $\lambda_0 = hc/\phi = 3\lambda/2$. Answer: C.

Question 2.4

A parallel beam of yellow light from a sodium discharge tube is directed to a glass tube filled with sodium vapour. Which of the following would happen after the sodium vapour absorbs the yellow light?

- A. No more yellow light can be seen.
- B. The sodium vapour emits yellow light in the direction of the incident beam.
- C. The sodium vapour emits yellow light in all directions.
- D. The sodium vapour emits white light in all directions.

Official answer: C

Sodium atoms absorb photons matching an allowed excitation energy. When the excited atoms return to lower levels, they re-emit the same characteristic yellow light in many directions, rather than only continuing the original beam direction. Answer: C.

Question 2.5

A beam of 8 keV electrons is directed towards a crystal to observe the diffraction of electrons. What is the de Broglie wavelength of an 8 keV electron?

- A. 4.34×10^{-10} m
- B. 1.37×10^{-11} m
- C. 1.74×10^{-19} m
- D. 5.49×10^{-21} m

Official answer: B

Using the non-relativistic approximation, $p = \sqrt{2m_e K}$ and $\lambda = h/p$. With $K = 8000(1.60 \times 10^{-19})$ J, $\lambda = (6.63 \times 10^{-34})/\sqrt{2(9.11 \times 10^{-31})(8000)(1.60 \times 10^{-19})} = 1.37 \times 10^{-11}$ m. Answer: B.

Question 2.6

The radio telescope situated in Guizhou province has an effective aperture of 300 m for observations. It can be used to observe electromagnetic waves of frequencies between 7×10^7 Hz and 3×10^9 Hz. Estimate the minimum angular separation that the telescope can resolve.

- A. 4.07×10^{-4} rad
- B. 9.49×10^{-4} rad
- C. 1.74×10^{-2} rad
- D. 4.07×10^{-2} rad

Official answer: A

For a circular aperture, $\theta_{\min} = 1.22\lambda/D$. Best resolution uses the shortest wavelength, corresponding to the highest frequency 3×10^9 Hz. Thus $\lambda_{\min} = c/f_{\max} = 0.10$ m and $\theta_{\min} = 1.22(0.10)/300 = 4.07 \times 10^{-4}$ rad. Answer: A.

Question 2.7

The leaves of the plant Edelweiss are covered with nanoscale filaments. These filaments absorb ultraviolet radiation but reflect all visible light. Which of the following statements are correct?

- (1) The leaves appear white in colour under sunlight as the filaments reflect all visible light.
- (2) Optical microscopes cannot be used to observe the filaments.
- (3) Due to their tiny size, the filaments do not pose a health risk if absorbed into the human body.

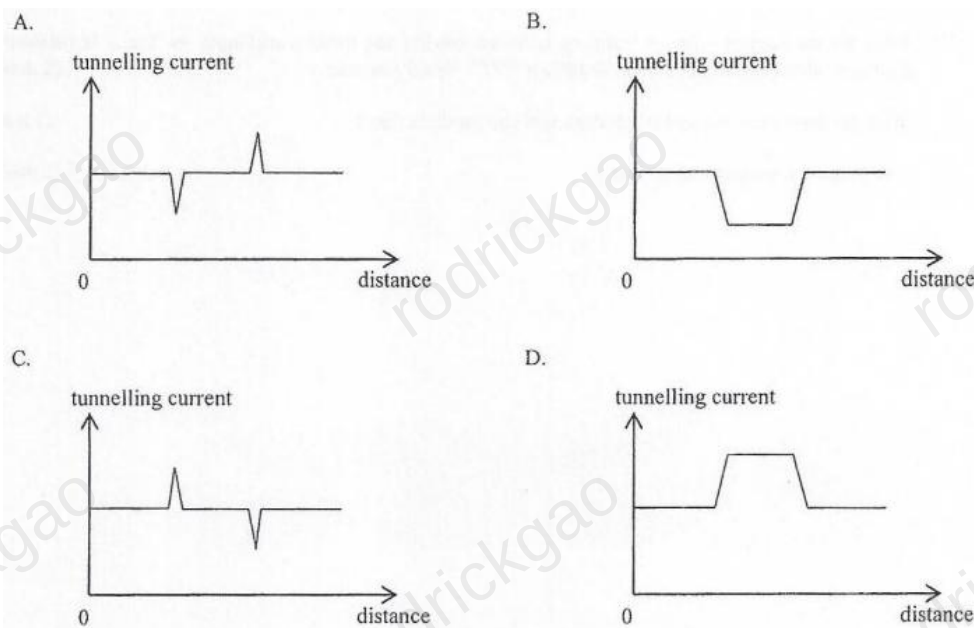
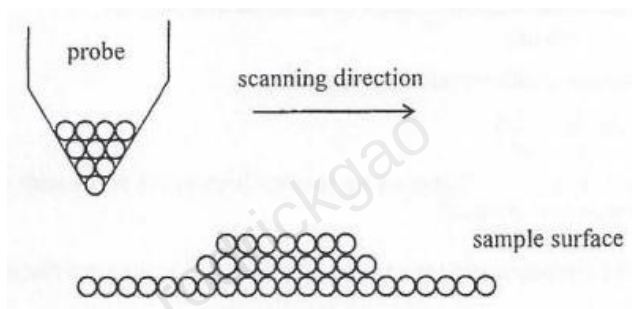
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: A

Reflecting all visible wavelengths makes the leaves appear white in sunlight. The nanoscale features are below the resolving ability of an ordinary optical microscope. Tiny size does not establish biological safety, so (3) is unjustified. In the intended scale comparison, (1) and (2) are correct. Answer: A.

Question 2.8

A scanning tunnelling microscope (STM) scans across a sample surface as shown. The probe scans horizontally across the sample surface at a fixed height. Which graph best represents the variation of the tunnelling current with distance travelled by the probe?



Official answer: D

At fixed probe height, the probe–surface gap becomes smaller over the raised part of the sample. Tunnelling current increases strongly as this gap decreases, remains high across the raised plateau, and falls again after it. The resulting raised-current plateau is shown in graph D. Answer: D.

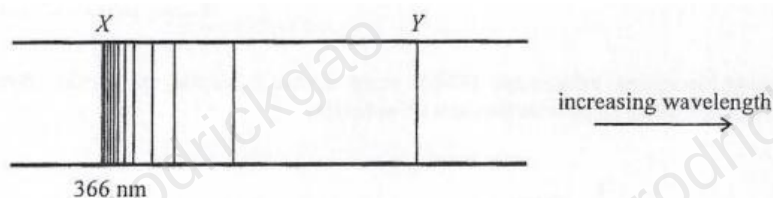
Q.2: Structured question

Figure 2.1 shows part of the line spectrum of hydrogen. It contains a series of spectral lines with wavelength λ given by

$$1/\lambda = R(1/2^2 - 1/n^2),$$

where R is a constant and $n = 3, 4, 5, \dots$. There are no spectral lines in the series with wavelength less than that of line X (366 nm) nor greater than that of line Y.

Figure 2.1



- (a) Use Bohr's model of the hydrogen atom to explain why the spectral lines are discrete but not continuous. (2 marks)
- (b)(i) Which region of the electromagnetic spectrum does line X belong to? (1 mark)
- (ii) What is the energy of a photon of line X? Express your answer in eV. (2 marks)
- (iii) What would happen when a beam of radiation having the same wavelength as line X is incident on hydrogen atoms in the first excited state ($n = 2$)? Briefly explain. (2 marks)
- (c)(i) State the transition in a hydrogen atom that can produce line Y. (1 mark)
- (ii) Determine the wavelength of line Y. (2 marks)

Official marking scheme

Solution		Marks	Remarks
2.	(a) When an atom transits from a higher energy level to a lower one, a photon with energy equals to the energy difference between the levels is emitted.	1A	
	Since energy levels are quantized, the energy (and thus wavelength) of the photons emitted can take some discrete values only.	1A	
		2	
	(b) (i) Line X belongs to the ultra-violet region.	1A	
		1	
	(ii) $\text{energy} = \frac{hc}{\lambda e}$ $= \frac{(6.63 \times 10^{-34})(3 \times 10^8)}{(366 \times 10^{-9})(1.60 \times 10^{-19})}$ $= 3.40 \text{ eV}$	1M	
		1A	
		2	
	(iii) The radiation would be absorbed, and the hydrogen atoms ionized.	1A	
		1A	
		2	
(c)	(i) The transition from $n = 3$ to $n = 2$. (i.e. from 2 nd to 1 st excited state)	1A	
		1	
	(ii) From line X, we have $\frac{1}{366} = R\left(\frac{1}{2^2} - 0\right)$ $R \approx 0.0109 \text{ (nm}^{-1}\text{)} \text{ (or } 1.09 \times 10^7 \text{ m}^{-1}\text{)}$ For line Y, $\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$ $\lambda = 658.8 \text{ nm}$	1M	
		1A	
	Alternative method: $R = \frac{13.6 \text{ eV}}{hc}$ $= \frac{13.6 \times (1.6 \times 10^{-19})}{(6.63 \times 10^{-34})(3 \times 10^8)}$ $= 1.094 \times 10^7 \text{ (m}^{-1}\text{)}$ $\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$ $\lambda = 6.58 \times 10^{-7} \text{ m}$ $E = E_2 - E_3$ $\frac{hc}{\lambda} = 13.6 \left(\frac{1}{2^2} - \frac{1}{3^2}\right) \text{ eV}$ $= 13.6 \left(\frac{1}{2^2} - \frac{1}{3^2}\right) \times 1.6 \times 10^{-19}$ $\lambda = 6.58 \times 10^{-7} \text{ m}$	1M	
		1A	
		2	

(a) Hydrogen has discrete allowed electron energies. A transition between two levels emits a photon with hf equal to their energy difference, so only particular photon energies and wavelengths occur.

(b)(i) Line X, at 366 nm, is ultraviolet.

(b)(ii) $E = hc/\lambda = (6.63 \times 10^{-34})(3.00 \times 10^8)/(366 \times 10^{-9}) = 5.43 \times 10^{-19} \text{ J} = 3.40 \text{ eV}$.

(b)(iii) X is the series limit, corresponding to $n = \infty \rightarrow 2$ in emission. In reverse, absorption of this photon supplies the ionization energy from $n = 2$, freeing the electron and ionizing the atom.

(c)(i) The longest-wavelength line has the smallest energy gap in this series: $n = 3 \rightarrow 2$.

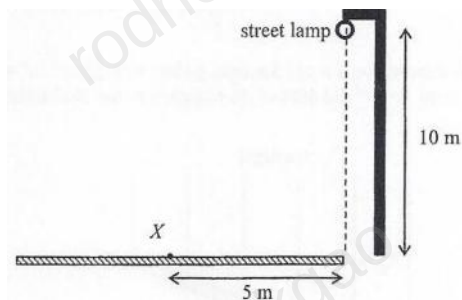
(c)(ii) At the limit, $1/\lambda_X = R/4$. For Y, $1/\lambda_Y = R(1/4 - 1/9) = 5R/36$. Hence $\lambda_Y = (9/5)\lambda_X = (9/5)(366) = 658.8 \text{ nm} \approx 659 \text{ nm}$. Use the supplied 366 nm limit consistently.

Section C: Energy and Use of Energy

Q.3: Multiple-choice questions

Question 3.1

A street lamp is installed 10 m above the ground on the side of a road. At a point X which is 5 m from the side on the road surface, the illuminance is 30 lux. Taking the lamp as a point source emitting light uniformly in all directions, and neglecting reflections and contributions from other light sources, estimate the luminous flux of the street lamp.



- A. $4.21 \times 10^4 \text{ lm}$
- B. $5.27 \times 10^4 \text{ lm}$
- C. $5.80 \times 10^4 \text{ lm}$
- D. $6.59 \times 10^4 \text{ lm}$

Official answer: B

The source-to-point distance is $r = \sqrt{(10^2 + 5^2)} = \sqrt{125} \text{ m}$, and $\cos\theta = 10/r$ for incidence on the horizontal ground. $E = [\Phi/(4\pi r^2)]\cos\theta$ gives $\Phi = 30(4\pi)(125)/(10/\sqrt{125}) = 5.27 \times 10^4 \text{ lm}$. The oblique-incidence factor must be included. Answer: B.

Question 3.2

Arrange the efficacy of the following light sources from the largest to the smallest.

Light source	X	Y	Z
Rated power / W	11	13	20
Luminous flux / lm	300	400	500

- A. X, Y, Z
- B. X, Z, Y
- C. Y, X, Z
- D. Y, Z, X

Official answer: C

Luminous efficacy is luminous flux divided by electrical power. X: $300/11 = 27.3 \text{ lm W}^{-1}$; Y: $400/13 = 30.8 \text{ lm W}^{-1}$; Z: $500/20 = 25.0 \text{ lm W}^{-1}$. Thus $Y > X > Z$, which is option C in the original paper. Answer: C.

Question 3.3

The surface of an induction cooker is usually made of strengthened glass. The glass surface gets hot after use mainly because

- A. an induction cooker has a high energy efficiency.
- B. heat is transferred from the cooking utensil to the glass surface when the cooking utensil gets hot.
- C. an eddy current flows in the glass.
- D. heat is produced in the solenoid of the cooker.

Official answer: B

The alternating magnetic field induces currents mainly in the conducting cooking utensil, heating it. Thermal energy then passes from the hot utensil to the glass by conduction. Glass does not support the substantial eddy currents responsible for cooking. Answer: B.

Question 3.4

Which of the following changes WILL NOT reduce the Overall Thermal Transfer Value (OTTV) of a building?

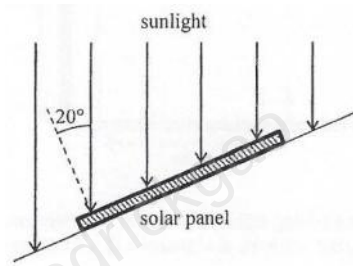
- A. building a roof-top garden
- B. installing heat insulating materials on the walls
- C. adding solar control window films to the windows
- D. painting the exterior of the building in dark colour

Official answer: D

A dark exterior absorbs more incident solar radiation, increasing heat gain rather than reducing OTTV in the stated comparison. Insulation, a roof garden and solar-control films reduce relevant heat-transfer contributions. Answer: D.

Question 3.5

A solar panel of area 3 m^2 is installed on a roof. Sunlight makes an angle of 20° to the normal of the panel at noon. The solar constant is 1366 W m^{-2} and 40% of the radiation power is absorbed by the atmosphere. If the efficiency of the solar panel is 10%, what is the electrical power generated by it at noon?



- A. 84 W
- B. 154 W
- C. 231 W
- D. 246 W

Official answer: C

Only 60% of the radiation remains after atmospheric absorption. The panel presents projected area $A \cos 20^\circ$ to the sunlight. Electrical power is $\eta(0.60)(1366)A \cos 20^\circ = 0.10(0.60)(1366)(3)\cos 20^\circ = 231 \text{ W}$. Answer: C.

Question 3.6

The figure shows a wind turbine. Which statements explain why the wind turbine is NOT 100% efficient in converting the kinetic energy of the wind to electrical energy?

- (1) There are mechanical energy losses in the moving parts.
- (2) Wind does not stop completely after passing through the rotor.
- (3) The direction of wind changes irregularly.



- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

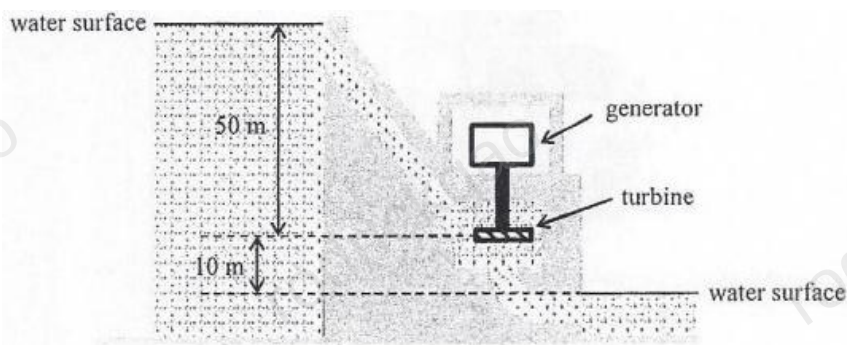
Official answer: D

Some energy is dissipated by moving parts. Wind must retain kinetic energy downstream for air to continue passing through the rotor. Changes of wind direction also reduce capture when the rotor is not aligned with the wind. All three reduce conversion of available wind energy to electrical output. Answer: D.

Question 3.7

The hydroelectric power plant shown has an efficiency of 40% in electricity generation. If the flow rate of the water is $300 \text{ m}^3 \text{ s}^{-1}$, what is the power output of the plant?

Given: density of water is 1000 kg m^{-3} . Take $g = 9.81 \text{ m s}^{-2}$.



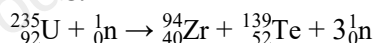
- A. 11.8 MW
- B. 58.9 MW
- C. 70.6 MW
- D. 88.3 MW

Official answer: Deleted

Official status: Deleted. The official marking scheme provides no accepted A–D key for this item. For supplementary practice only, $P = \eta \rho Q g H$: taking $H = 50 \text{ m}$ gives 58.9 MW, while taking the full 60 m difference between water surfaces gives 70.6 MW. These illustrate sensitivity to the hydraulic-head interpretation; neither is presented here as an official answer, and the marking scheme does not state the reason for deletion.

Question 3.8

Energy is released in the following nuclear fission of uranium-235:



Which statements concerning the reaction are correct?

- (1) The rate of the reaction can be controlled by absorbing some of the neutrons produced.
- (2) Mass is conserved in the reaction.
- (3) The binding energy per nucleon of ${}_{92}^{235}\text{U}$ is higher than that of ${}_{40}^{94}\text{Zr}$ or ${}_{52}^{139}\text{Te}$.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: A

Absorbing some released neutrons controls how many further fissions occur, so (1) is correct. A decrease of rest mass supplies the released energy, so rest mass alone is not conserved. The products are more tightly bound per nucleon than U-235, not less. Answer: A.

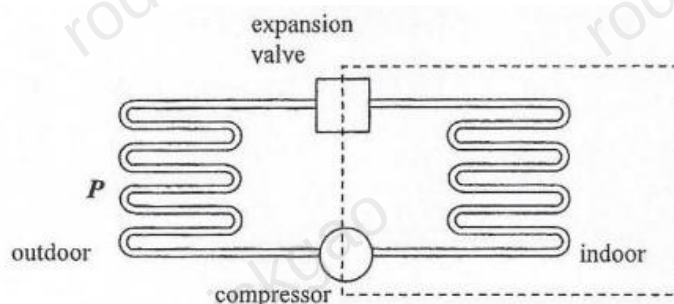
Q.3: Structured question

A refrigerated truck is used for transporting frozen goods. A refrigerator is installed in the refrigerated compartment.



(a) Figure 3.1 shows a simplified schematic diagram of a refrigerator.

Figure 3.1

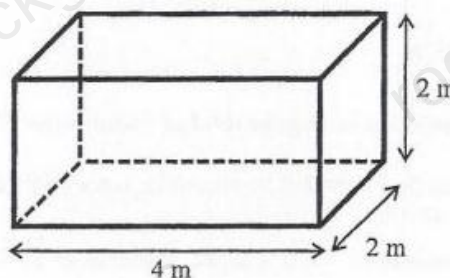


(i) In which direction does the refrigerant flow through the compressor (from indoor to outdoor or from outdoor to indoor)? (1 mark)

(ii) Describe the change of state of the refrigerant and the heat exchange when it flows through component P. (2 marks)

(b) Figure 3.2 shows the dimensions of the refrigerated compartment. The compartment is insulated using 0.08 m thick polystyrene. The thermal conductivity of polystyrene is $0.03 \text{ W m}^{-1} \text{ K}^{-1}$.

Figure 3.2



(i) If a temperature difference of 50°C is maintained between the exterior and the interior surfaces, estimate the minimum cooling capacity required for the refrigerator. (Hint: consider all the surfaces of the compartment.) (3 marks)

(ii) On a sunny afternoon, the AIR TEMPERATURE is 35°C . By using the refrigerator with cooling capacity calculated in (b)(i), briefly explain why the temperature inside the compartment CANNOT be maintained at -15°C . (2 marks)

(c) Light emitting diodes (LED) are installed inside the refrigerated compartment for illumination. State TWO advantages of using LED over other common types of lighting. (2 marks)

Official marking scheme

Solution		Marks	Remarks	
3.	(a) (i)	The refrigerant flows from indoor to outdoor through the compressor.	1A	
			1	
	(ii)	The refrigerant condenses / changes from gas to liquid. It releases the heat/internal energy to the environment.	1A 	
			1A	
	(b) (i)	Total surface area = $(4 \times 2) \times 4 + (2 \times 2) \times 2 = 40 \text{ m}^2$	1M	
		Cooling capacity = rate of heat gain		
		$= \kappa \frac{A(T_H - T_C)}{d} = 0.03 \frac{40(50)}{0.08}$	1M	
		$= 750 \text{ W}$	1A	
			3	
		(ii)	The compartment absorbs heat by radiation, the exterior surface temperature of the refrigerated compartment is higher than 35°C . Thus, the interior will be higher than -15°C .	1A
				1A
	(c)	Light emitting diode (LED) has a long life time and very high efficacy.	2	
			1A	
			1A	
			2	

(a)(i) Through the compressor, refrigerant flows from indoor to outdoor. The compressor draws vapour from the indoor evaporator and sends it towards the outdoor condenser.

(a)(ii) In component P, refrigerant condenses from gas to liquid and releases thermal energy to the surroundings.

(b)(i) Include all six faces: $A = 4(4 \times 2) + 2(2 \times 2) = 40 \text{ m}^2$. The minimum steady cooling capacity equals conductive heat gain, $P = \kappa A \Delta T / d = 0.03(40)(50) / 0.08 = 750 \text{ W}$.

(b)(ii) Sunlight heats the exterior surface above the air temperature of 35°C . Maintaining -15°C inside would then require a surface temperature difference greater than 50°C and heat removal above 750 W . The calculated refrigerator capacity is insufficient.

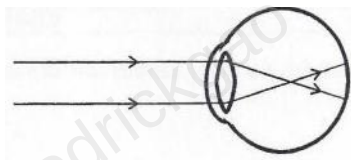
(c) LEDs have high luminous efficacy, so less electrical power is needed for the required light, and they have a long service life, reducing replacement frequency.

Section D: Medical Physics

Q.4: Multiple-choice questions

Question 4.1

The figure shows an eye looking at a distant object. Which of the following is correct?



- A. Lens of the eye too thick; corrected by spectacles with a diverging lens
- B. Lens of the eye too thick; corrected by spectacles with a converging lens
- C. Lens of the eye too thin; corrected by spectacles with a diverging lens
- D. Lens of the eye too thin; corrected by spectacles with a converging lens

Official answer: A

Parallel rays from a distant object focus in front of the retina: the eye has too much converging power in the illustrated model. A diverging spectacle lens reduces the incoming convergence so the image can form on the retina. Answer: A.

Question 4.2

Which of the following statements about human hearing are correct?

- (1) The ear bones in the middle ear convert sound waves into vibrations of the ear drum.
- (2) Pressure is amplified because of the difference in area between the ear drum and the oval window.
- (3) Mechanical vibrations are converted into electrical signals in the inner ear.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: C

Sound pressure first vibrates the eardrum; the ear bones transmit this vibration, not the reverse process stated in (1). The smaller area of the oval window contributes to pressure amplification. Sensory cells in the inner ear convert mechanical vibration to electrical nerve signals. Hence (2) and (3) are correct. Answer: C.

Question 4.3

The sound intensity level in a factory is 95 dB. The workers wear ear protectors that can reduce the sound intensity level by 30 dB. What is the intensity of sound as heard by the workers?

Given: threshold of hearing $I_0 = 1 \times 10^{-12} \text{ W m}^{-2}$.

- A. $1.00 \times 10^{-9} \text{ W m}^{-2}$
- B. $3.16 \times 10^{-6} \text{ W m}^{-2}$
- C. $3.16 \times 10^{-3} \text{ W m}^{-2}$
- D. 3.16 W m^{-2}

Official answer: B

The protected level is $95 - 30 = 65 \text{ dB}$. Using $\beta = 10 \log_{10}(I/I_0)$, $I = I_0 \times 10^{\beta/10} = 10^{-12} \times 10^{6.5} = 3.16 \times 10^{-6} \text{ W m}^{-2}$. A 30 dB reduction divides intensity by 1000; it does not subtract 30 from the intensity. Answer: B.

Question 4.4

The acoustic impedances of various tissues and that of air are listed in the table. Which interface will give the largest intensity reflection coefficient in ultrasound scans?

	acoustic impedance ($\times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$)
fat	1.34
liver	1.65
muscle	1.71
bone	7.8
air	0.0004

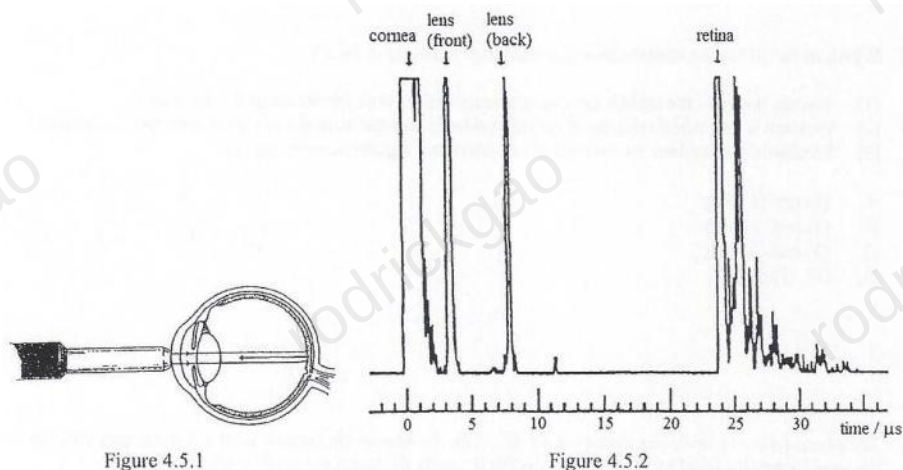
- A. liver–muscle
- B. fat–muscle
- C. muscle–bone
- D. muscle–air

Official answer: D

At normal incidence the intensity reflection coefficient is $[(Z_2 - Z_1)/(Z_2 + Z_1)]^2$. The air–muscle impedance mismatch is proportionally the largest: $[(1.71 - 0.0004)/(1.71 + 0.0004)]^2 \approx 0.999$, exceeding the other listed interfaces. Answer: D.

Question 4.5

An ultrasound transducer is used to scan the eye and the echoes received are shown. The velocity of the ultrasound waves in the eye is 1550 m s^{-1} . The thickness of the lens is about



- A. 1.6 mm
- B. 3.5 mm**
- C. 7.0 mm
- D. 18.6 mm

Official answer: B

The front and back lens echoes are separated by about $4.5 \mu\text{s}$. This delay includes travel through the lens and back, so thickness $d = v\Delta t/2 = 1550(4.5 \times 10^{-6})/2 \approx 3.5 \times 10^{-3} \text{ m} = 3.5 \text{ mm}$. Do not use the time from the initial pulse or omit the factor of two. Answer: B.

Question 4.6

Which of the following statements about an endoscope is correct?

- A. Coherent bundle fibres are used to transmit image.**
- B. Light can only travel from the objective to the eyepiece, but not in the opposite direction.
- C. The refractive index of the cladding of the glass fibre is higher than that of glass.
- D. An endoscope can only show black-and-white images.

Official answer: A

A coherent fibre bundle preserves the relative arrangement of its fibres and therefore transmits an image. Light paths are reversible, the core must have higher refractive index than the cladding for total internal reflection, and visible colours can be transmitted. Answer: A.

Question 4.7

A certain tracer Y has a biological half-life of 3 days and a physical half-life of 4 hours. What is the effective half-life of Y?

- A. 0.24 hours
- B. 1.71 hours
- C. 3.79 hours
- D. 4.23 hours

Official answer: C

Convert the biological half-life to 72 h. Physical decay and biological removal act together: $1/T_{\text{eff}} = 1/72 + 1/4$.
Hence $T_{\text{eff}} = 72 \times 4 / (72 + 4) = 3.79$ h, shorter than either individual half-life. Answer: C.

Question 4.8

Which of the following statements about radionuclide imaging is correct?

- A. Due to the decay of the tracer, images should be taken immediately after the tracer is injected.
- B. The gamma camera emits gamma radiation to irradiate the tracer.
- C. Radionuclide imaging can clearly reveal the structure of a failed organ.
- D. For a period of time after injecting the tracer, excretion of the patient may be radioactive.

Official answer: D

Some tracer is removed biologically before all of it decays, so excretion may remain radioactive for a time. A gamma camera detects photons emitted by the tracer; it does not irradiate it. Radionuclide imaging primarily maps tracer uptake and function, and imaging time must allow suitable distribution. Answer: D.

Q.4: Structured question

X-ray radiographic imaging and computed tomography (CT) scans are used for medical purposes.

- (a) Briefly describe how X-ray is produced. (1 mark)
- (b) State an advantage of a CT scan over X-ray radiographic imaging. (1 mark)
- (c) The effective dose of radiation absorbed can be measured in millisieverts (mSv) or expressed as the time taken to receive the equivalent dose from background radiation. The effective doses for a chest X-ray radiographic imaging and a chest CT scan are shown below.

Imaging method	Effective dose / mSv	Equivalent background dose / days
Chest X-ray radiographic imaging	0.02	1.85
Chest CT scan	6.6	610.5

- (i) Briefly explain why the effective dose of a CT scan is much higher. (1 mark)
- (ii) A head CT scan has an effective dose of 1.5 mSv. Based on the information from the table, estimate its equivalent background radiation dose. (1 mark)
- (d) In a CT scan, a narrow X-ray beam of initial intensity I_0 transmits through lung cavity, soft tissue and bone along its path. The table below shows the linear attenuation coefficients and path lengths.

Tissue	Linear attenuation coefficient / cm^{-1}	Path length / cm
Lung cavity	0.1	19.8
Soft tissue	0.18	8.8
Bone	0.48	4.4

- (i) Briefly explain the large difference in linear attenuation coefficient between lung cavity and bone. (1 mark)
- (ii) Determine the value of transmitted intensity I / initial intensity I_0 of the X-ray after transmitted through lung cavity, soft tissue and bone. (3 marks)
- (e) A student suggests that a CT scan can be used for checking a foetus. Briefly explain whether you agree or not. If you do not agree, suggest a suitable medical imaging method for checking a foetus. (2 marks)

Official marking scheme

Solution	Marks	Remarks
4. (a) X-ray is produced when fast electrons hit a heavy metal target.	1A	
	1	
(b) CT scan is better at mapping soft tissues / differentiating between overlying structures in the body / making 3D images	1A	
	1	
(c) (i) The effective dose of CT scan is much higher because multiple X-ray images are taken for a CT scan.	1A	
	1	
(ii) Equivalent background radiation dose $= 1.85 \times \frac{1.5}{0.02}$ $= 138.75 \text{ days}$	1A	
	1	
(d) (i) The lung cavity is filled with air / There is a large difference in density between the lung cavity and bone	1A	
	1	
(ii) $I = I_0 e^{-(\mu_1 x_1 + \mu_2 x_2 + \mu_3 x_3)}$ $\frac{I}{I_0} = e^{-(0.1 \times 19.8 + 0.18 \times 8.8 + 0.48 \times 4.4)}$ $= e^{-5.676} = 3.43 \times 10^{-3}$	1M+1M 1A	
	3	
(e) I do not agree because a CT scan may cause ionization (changes) in cells / damage DNA of the foetus. An ultrasound scan can be used for checking a foetus.	1A 1A	
	2	

(a) Fast electrons strike a heavy-metal target and are rapidly decelerated, producing X-rays.

(b) CT reconstructs cross-sectional images, reducing overlap of structures and allowing better separation of internal tissues than a single projection radiograph.

(c)(i) A CT scan acquires many X-ray projections from different directions, giving a larger cumulative exposure in this comparison.

(c)(ii) Equivalent background time = $1.85 \times (1.5/0.02) = 138.75 \text{ days} \approx 139 \text{ days}$.

(d)(i) The lung cavity contains low-density air, whereas bone is much denser and attenuates X-rays more strongly. Their compositions also differ.

(d)(ii) Multiply the transmission factors for successive layers: $I/I_0 = e^{-\mu_1 x_1} e^{-\mu_2 x_2} e^{-\mu_3 x_3} = e^{-0.1 \times 19.8 - 0.18 \times 8.8 - 0.48 \times 4.4} = e^{-5.676} = 3.43 \times 10^{-3}$, about 0.343%. Use centimetres with the coefficients in cm^{-1} .

(e) For the proposed routine foetal check, disagree: CT uses ionizing radiation that can damage developing cells. Ultrasound is the suitable non-ionizing imaging method in this question.